

Linear Equations & Systems



Solving single-variable linear equations

- 1 Solve for x : $3x - 7 = 20$.
 - 2 Solve for x : $\frac{2x + 5}{3} = 7$.
 - 3 Solve for x : $5(x - 2) = 3(x + 4)$.
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Equations with no solution or infinitely many solutions

- 4 For what value of k does the equation $4x + 6 = kx + 6$ have infinitely many solutions?
 - 5 Explain why the equation $2x + 3 = 2x + 7$ has no solution, without solving it algebraically first.
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Solving systems by substitution

- 6 Solve the system: $y = 2x + 1$ and $3x + y = 16$.
 - 7 Solve the system: $x - 2y = 5$ and $2x + 3y = 3$.
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Solving systems by elimination

- 8 Solve the system: $2x + y = 11$ and $x - y = 1$, using elimination.
 - 9 Solve the system: $3x + 2y = 16$ and $5x - 2y = 8$, using elimination.
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Systems with no solution or infinitely many solutions

- 10 For what value of c does the system $6x + 9y = 12$ and $2x + 3y = c$ have infinitely many solutions?
 - 11 Explain why the system $4x - 2y = 10$ and $2x - y = 3$ has no solution.
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Translating word problems into linear equations

- 12 A rectangle's length is 4 more than twice its width. If the perimeter is 68, find the width and length.
- 13 A taxi charges a flat fee of 4 plus 2.50 per mile. If a ride costs 21.50, how many miles was the ride?
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Systems as word problems

- 14 A movie theater sells adult tickets for 12 and child tickets for 8. If 150 tickets were sold for a total of 1560, how many of each type were sold?
- 15 Two numbers have a sum of 45 and a difference of 11. Find the two numbers.
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Interpreting the slope and intercept of a linear model

- 16 A gym membership costs $C = 25 + 15m$ dollars, where m is the number of months. Interpret the meaning of the numbers 25 and 15 in this context.
- 17 Two companies charge for printing: Company A charges $C = 0.10p + 20$ and Company B charges $C = 0.15p$, where p is the number of pages. Find the number of pages at which both companies charge the same amount, and state which company is cheaper for 500 pages.
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Equations with variables on both sides and fractions

- 18 Solve for x : $\frac{x}{3} + \frac{x}{4} = 7$.
- 19 Solve for x : $2(x + 3) - 5 = 3(x - 1) + 2$.
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Systems with three unknowns represented as a word problem

- 20 A vending machine sells snacks for 1.50 dollars and drinks for 2 dollars. On one day it sold 45 items total for 79.50 dollars. Set up and solve a system to find the number of snacks and drinks sold.
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Choosing the best method for a given system

- 21 For the system $y = 4x - 3$ and $5x + 2y = 17$, explain which method (substitution or elimination) is more efficient here, and solve the system.
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Consistent, inconsistent, and dependent systems

- 22 Classify the system $y = 3x + 2$ and $y = 3x - 4$ as consistent, inconsistent, or dependent, and explain your reasoning without solving.